

Web Appendices for “Covariate-adjusted win statistics in randomized clinical trials with ordinal outcomes”

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This documentation presented the web appendices and tables not shown in the manuscript.

Web Appendix A: Consistent estimators for variance of unadjusted estimator

A1. Estimation of variance for win probability and loss probability

For unadjusted estimator $\hat{\tau}_1^{\text{UNADJ}}$, we have

$$\hat{\tau}_1^{\text{UNADJ}} = \frac{1}{n_1 n_0} \sum_{i=1}^{n_1} \sum_{j=1}^{n_0} \omega_{ij} \mathbb{I}(Y_i > Y_j) = \frac{\sum_{i=1}^n \sum_{j=1}^n Z_i (1 - Z_j) \mathbb{I}(Y_i > Y_j)}{\sum_{i=1}^n \sum_{j=1}^n Z_i (1 - Z_j)},$$

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where $\omega_{ij} = \frac{Z_i(1-Z_j)}{\frac{1}{n_1} \sum_{i=1}^{n_1} Z_i \times \frac{1}{n_0} \sum_{j=1}^{n_0} (1-Z_j)}$. In main text, we have derived its asymptotic variance as $\sigma_1^2 = \frac{n}{n_1} \xi_{10} + \frac{n}{n_0} \xi_{01}$, $n = n_1 + n_0$ and

$$\begin{aligned}\xi_{10} &= \text{cov}(\omega_{ij}\mathbb{I}(Y_i > Y_j), \omega_{ik}\mathbb{I}(Y_i > Y_k)) \\ &= E[\omega_{ij}\mathbb{I}(Y_i > Y_j) \times \omega_{ik}\mathbb{I}(Y_i > Y_k)] - E[\omega_{ij}\mathbb{I}(Y_i > Y_j)]E[\omega_{ik}\mathbb{I}(Y_i > Y_k)] \\ &= P(Y_i(1) > Y_j(0) \& Y_i(1) > Y_k(0)) - [P(Y_i(1) > Y_j(0))]^2, \\ \xi_{01} &= \text{cov}(\omega_{ij}\mathbb{I}(Y_i > Y_j), \omega_{kj}\mathbb{I}(Y_k > Y_j)), \\ &= E[\omega_{ij}\mathbb{I}(Y_i > Y_j) \times \omega_{kj}\mathbb{I}(Y_k > Y_j)] - E[\omega_{ij}\mathbb{I}(Y_i > Y_j)]E[\omega_{kj}\mathbb{I}(Y_k > Y_j)] \\ &= P(Y_i(1) > Y_j(0) \& Y_k(1) > Y_j(0)) - [P(Y_i(1) > Y_j(0))]^2, \quad i, j, k = 1, \dots, n.\end{aligned}$$

To obtain consistent estimators, for σ_1^2 , similar to the approach of Bebu and Lachin (2016), we denote $S_{i\cdot}^1 = \sum_{j=1}^{n_0} \omega_{ij}\mathbb{I}(Y_i > Y_j)$, $S_{\cdot j}^2 = \sum_{i=1}^{n_0} \omega_{ij}\mathbb{I}(Y_i < Y_j)$, $S_{i\cdot}^1 = \sum_{i=1}^{n_1} \omega_{ij}\mathbb{I}(Y_i > Y_j)$ and $S_{\cdot j}^2 = \sum_{i=1}^{n_1} \omega_{ij}\mathbb{I}(Y_i < Y_j)$. Then the above probabilities can be estimated as follows:

$$\begin{aligned}\hat{P}(Y_i(1) > Y_j(0)) &= \hat{E}[\mathbb{I}(Y_i(1) > Y_j(0))] = \hat{\tau}_1^{\text{UNADJ}} = \frac{\sum_{i=1}^n \sum_{j=1}^n Z_i(1-Z_j)\mathbb{I}(Y_i > Y_j)}{\sum_{i=1}^n \sum_{j=1}^n Z_i(1-Z_j)}, \\ \hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) > Y_k(0)) &= \frac{\sum_{i=1}^{n_1} S_{i\cdot}^1(S_{i\cdot}^1 - 1)}{n_1 n_0 (n_0 - 1)} \\ &= \frac{\sum_{i=1}^{n_1} \left\{ \sum_{j=1}^{n_0} \omega_{ij}\mathbb{I}(Y_i > Y_j) [\sum_{j=1}^{n_0} \omega_{ij}\mathbb{I}(Y_i > Y_j) - 1] \right\}}{n_1 n_0 (n_0 - 1)}, \\ &= \frac{\sum_{i=1}^{n_1} \left[\sum_{j=1}^{n_0} \omega_{ij}\mathbb{I}(Y_i > Y_j) \right]^2}{n_1 n_0 (n_0 - 1)} - \frac{1}{n_0 - 1} \hat{P}(Y_i(1) > Y_j(0)), \\ \hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) > Y_j(0)) &= \frac{\sum_{j=1}^{n_0} S_{\cdot j}^1(S_{\cdot j}^1 - 1)}{n_0 n_1 (n_1 - 1)} \\ &= \frac{\sum_{j=1}^{n_0} \left\{ \sum_{i=1}^{n_1} \omega_{ij}\mathbb{I}(Y_i > Y_j) [\sum_{i=1}^{n_1} \omega_{ij}\mathbb{I}(Y_i > Y_j) - 1] \right\}}{n_0 n_1 (n_1 - 1)} \\ &= \frac{\sum_{j=1}^{n_0} \left[\sum_{i=1}^{n_1} \omega_{ij}\mathbb{I}(Y_i > Y_j) \right]^2}{n_0 n_1 (n_1 - 1)} - \frac{1}{n_1 - 1} \hat{P}(Y_i(1) > Y_j(0)).\end{aligned}$$

Therefore, a consistent variance estimator for $\hat{\tau}_1^{\text{UNADJ}}$ is

$$\begin{aligned}\widehat{\text{var}}(\hat{\tau}_1^{\text{UNADJ}}) &= \frac{1}{n} \left\{ \frac{n}{n_1} \hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) > Y_k(0)) - \frac{n}{n_1} [\hat{P}(Y_i(1) > Y_j(0))]^2 \right. \\ &\quad \left. + \frac{n}{n_0} \hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) > Y_j(0)) - \frac{n}{n_0} [\hat{P}(Y_i(1) > Y_j(0))]^2 \right\} \\ &= \frac{1}{n_1} \hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) > Y_k(0)) + \frac{1}{n_0} \hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) > Y_j(0)) - \frac{n}{n_1 n_0} [\hat{P}(Y_i(1) > Y_j(0))]^2.\end{aligned}$$

By replacing operator “>” by “<” in the above estimates, i.e., replacing “>” in $\hat{P}(Y_i(1) > Y_j(0))$, $\hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) > Y_k(0))$ and $\hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) > Y_j(0))$ by “<”, we can obtain $\widehat{\text{var}}(\hat{\tau}_{-1}^{\text{UNADJ}})$, where $\hat{\tau}_{-1}^{\text{UNADJ}} = \frac{\sum_{i=1}^n \sum_{j=1}^n Z_i(1-Z_j)\mathbb{I}(Y_i < Y_j)}{\sum_{i=1}^n \sum_{j=1}^n Z_i(1-Z_j)}$.

A2. Estimation of covariance between win probability and loss probability

For the asymptotic covariance $\sigma_{12} = \frac{n}{n_1}\xi_{10}^{12} + \frac{n}{n_0}\xi_{01}^{12}$, where

$$\begin{aligned}
\xi_{10}^{12} &= \text{cov}(\omega_{ij}\mathbb{I}(Y_i > Y_j), \omega_{ik}\mathbb{I}(Y_i < Y_k)), \\
&= E[\omega_{ij}\mathbb{I}(Y_i > Y_j) \times \omega_{ik}\mathbb{I}(Y_i < Y_k)] - E[\omega_{ij}\mathbb{I}(Y_i > Y_j)]E[\omega_{ik}\mathbb{I}(Y_i < Y_k)] \\
&= P(Y_i(1) > Y_j(0) \& Y_i(1) < Y_k(0)) - P(Y_i(1) > Y_j(0))P(Y_i(1) < Y_k(0)), \\
\xi_{01}^{12} &= \text{cov}(\omega_{ij}\mathbb{I}(Y_i > Y_j), \omega_{kj}\mathbb{I}(Y_k < Y_j)), \\
&= E[\omega_{ij}\mathbb{I}(Y_i > Y_j) \times \omega_{kj}\mathbb{I}(Y_k < Y_j)] - E[\omega_{ij}\mathbb{I}(Y_i > Y_j)]E[\omega_{kj}\mathbb{I}(Y_k < Y_j)] \\
&= P(Y_i(1) > Y_j(0) \& Y_k(1) < Y_j(0)) - P(Y_i(1) > Y_j(0))P(Y_k(1) < Y_j(0)).
\end{aligned}$$

Similarly, we can obtain their consistent estimates as

$$\begin{aligned}
\hat{P}(Y_i(1) < Y_j(0)) &= \frac{\sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)\mathbb{I}(Y_i < Y_j)}{\sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)}, \\
\hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) < Y_k(0)) &= \frac{\sum_{i=1}^{n_1} S_{i\cdot}^1 S_{i\cdot}^2}{n_1 n_0 (n_0 - 1)} \\
&= \frac{\sum_{i=1}^{n_1} \left\{ \sum_{j=1}^{n_0} \omega_{ij}\mathbb{I}(Y_i > Y_j) [\sum_{j=1}^{n_0} \omega_{ij}\mathbb{I}(Y_i < Y_j)] \right\}}{n_1 n_0 (n_0 - 1)}, \\
\hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) < Y_j(0)) &= \frac{\sum_{j=1}^{n_0} S_{\cdot j}^1 S_{\cdot j}^2}{n_0 n_1 (n_1 - 1)} \\
&= \frac{\sum_{j=1}^{n_0} \left\{ \sum_{i=1}^{n_1} \omega_{ij}\mathbb{I}(Y_i > Y_j) [\sum_{i=1}^{n_1} \omega_{ij}\mathbb{I}(Y_i < Y_j)] \right\}}{n_0 n_1 (n_1 - 1)}.
\end{aligned}$$

Therefore, the covariance estimation between $\hat{\tau}_1^{\text{UNADJ}}$ and $\hat{\tau}_{-1}^{\text{UNADJ}}$ is $\frac{1}{n}\hat{\sigma}_{12}$, which can be computed as

$$\begin{aligned}
\text{cov}(\hat{\tau}_1^{\text{UNADJ}}, \hat{\tau}_{-1}^{\text{UNADJ}}) &= \frac{1}{n} \left\{ \frac{n}{n_1} \hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) < Y_k(0)) - \frac{n}{n_1} \hat{P}(Y_i(1) > Y_j(0)) \hat{P}(Y_i(1) < Y_k(0)) \right. \\
&\quad \left. + \frac{n}{n_0} \hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) < Y_j(0)) - \frac{n}{n_0} \hat{P}(Y_i(1) > Y_j(0)) \hat{P}(Y_k(1) < Y_j(0)) \right\} \\
&= \frac{1}{n_1} \hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) < Y_k(0)) + \frac{1}{n_0} \hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) < Y_j(0)) \\
&\quad - \frac{n}{n_1 n_0} \hat{P}(Y_i(1) > Y_j(0)) \hat{P}(Y_i(1) < Y_k(0)).
\end{aligned}$$

Web Appendix B: Proof for Theorem 1

B1. Proof for Theorem 1(a)

Note that the stabilized weighting estimator $\hat{\tau}_1^h$ defined in section 3.3 of main text, that is,

$$\hat{\tau}_1^h = \binom{n}{2}^{-1} \sum_{i=1}^n \sum_{j>i}^n q^h(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta}),$$

is a U-statistic, where $q^h(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta}) = \frac{1}{2} [Z_i(1 - Z_j)\mathbb{I}(Y_i > Y_j)\omega(X_i, X_j) + Z_j(1 - Z_i)\mathbb{I}(Y_i < Y_j)\omega(X_j, X_i)] / \left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)\omega(X_i, X_j) \right]$. For balanced weighting $\omega(\cdot, \cdot)$, $q^h(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta})$ is a real-valued symmetric

function. It is easy to see that $(E \times E)q^h(\cdot, \cdot; \beta)^2 < \infty$ and the function $q^h(\mathcal{O}_i, \mathcal{O}_j; \beta)$ is continuously differentiable with respect to β on Φ_δ , where $\Phi_\delta = \{\beta \in \Phi : \|\beta - \beta_*\| < \delta\}$ with β_* as the true value of β , and Φ is the range of β with $\|\cdot\|$ denotes the Euclidean norm and $\delta > 0$.

Therefore, by the Lemma S1 of Mao (2018), it is clear that the influence function of $\hat{\tau}_1^h$ is

$$\phi(\mathcal{O}) = 2\{q_1^h(\mathcal{O}; \beta) - \tau_1^h\} + \frac{\partial}{\partial \beta^T}(\mathbb{E} \times \mathbb{E})q^h(\mathcal{O}, \mathcal{O}; \beta)\tilde{l}_\beta(\mathcal{O}; \beta), \quad (\text{A.1})$$

where $q_1^h(\mathcal{O}; \beta) = \mathbb{E}q^h(\cdot, \mathcal{O}; \beta) = \frac{1}{2} \left[\frac{Z}{e(X)}g_1(Y) + \frac{1-Z}{1-e(X)}g_0(Y) \right]$ with $g_1(Y) = \int \mathbb{I}(Y > s)\nu_0(ds)$ and $g_0(Y) = \int \mathbb{I}(Y < t)\nu_1(dt)$, and $\tilde{l}_\beta(\mathcal{O}; \beta)$ is the influence function for β . Since $\hat{\tau}_1^h$ is consistent with τ_1^h , the asymptotic variance of $n^{1/2}(\hat{\tau}_1^h - \tau_1^h)$ is $\text{var}(\phi(\mathcal{O}))$.

B2. Proof for Theorem 1(b)

To show $\phi(\mathcal{O}) = \chi^h(\mathcal{O}) - \prod\{\chi^h(\mathcal{O})|\Lambda_e\}$, where

$$\chi^h(\mathcal{O}) = \frac{Z}{e(X)}g_1(Y) + \frac{1-Z}{1-e(X)}g_0(Y) - 2\tau_1^h,$$

and $\prod\{\chi^h(\mathcal{O})|\Lambda_e\}$ is the projection of $\chi^h(\mathcal{O})$ onto Λ_e , the tangent space of $\mathcal{M}_e$. Note that according to Tsiatis (2006), we can write Λ_e and $\prod\{\chi^h(\mathcal{O})|\Lambda_e\}$ as

$$\Lambda_e = \{B^{1 \times p}S_\beta(\mathcal{O}; \beta) \mid \forall \text{ real matrices } B^{1 \times p}\}$$

and

$$\prod\{\chi^h(\mathcal{O})|\Lambda_e\} = \mathbb{E}(\chi^h(\mathcal{O})S_\beta^T) \{\mathbb{E}(S_\beta S_\beta^T)\}^{-1} S_\beta(\mathcal{O}; \beta),$$

respectively, where $S_\beta(\mathcal{O}; \beta)$ is the efficient score function of β . To see this, observe that $\phi(\mathcal{O}) = \chi^h(\mathcal{O}) - \prod\{\chi^h(\mathcal{O})|\Lambda_e\}$ is equivalent to

$$\frac{\partial}{\partial \beta^T}(E \times E)q^h(\mathcal{O}, \mathcal{O}; \beta)\tilde{l}_\beta(\mathcal{O}; \beta) = -\prod\{\chi^h(\mathcal{O})|\Lambda_e\}. \quad (\text{A.2})$$

Note that the influence function $\tilde{l}_\beta(\mathcal{O}; \beta) = \{ES_\beta(\mathcal{O}; \beta)^{\otimes 2}\}^{-1} S_\beta(\mathcal{O}; \beta)$ and

$$\frac{\partial}{\partial \beta^T}(E \times E)q^h(\mathcal{O}, \mathcal{O}; \beta)\tilde{l}_\beta(\mathcal{O}; \beta) = \tilde{l}_\beta^T(\mathcal{O}; \beta) \frac{\partial}{\partial \beta}(E \times E)q^h(\mathcal{O}, \mathcal{O}; \beta).$$

So if we can show

$$\frac{\partial}{\partial \beta}(E \times E)q^h(\mathcal{O}, \mathcal{O}; \beta) = -2E\{q_1^h(\mathcal{O}; \beta) - \tau_1^h\} S_\beta(\mathcal{O}; \beta), \quad (\text{A.3})$$

(A.2) is straightforwardly proved since $2[q_1^h(\mathcal{O}; \beta) - \tau_1^h] = \chi^h(\mathcal{O})$.

To prove (A.3), write the density of $\mathcal{O}$ as $f(x; \beta)$ with respect to some dominating measure κ when the true parameter in $\mathcal{M}_e$ is β . Then for all β , we have

$$\int \int \{q^h(x_1, x_2; \beta) - \tau_1^h\} f(x_1; \beta) f(x_2; \beta) \kappa(dx_1) \kappa(dx_2) = 0.$$

Taking the derivative of both sides with respect to β , we obtain

$$\begin{aligned}
0 &= \frac{\partial}{\partial \beta} \int \int \{q^h(x_1, x_2; \beta) - \tau_1^h\} f(x_1; \beta) f(x_2; \beta) \kappa(dx_1) \kappa(dx_2) \\
&= \int \int \frac{\partial}{\partial \beta} [\{q^h(x_1, x_2; \beta) - \tau_1^h\} f(x_1; \beta) f(x_2; \beta) \kappa(dx_1) \kappa(dx_2)] \\
&= \frac{\partial}{\partial \beta} (E \times E) q^h(\cdot, \cdot; \beta) + \int \int \{q^h(x_1, x_2; \beta) - \tau_1^h\} \dot{f}(x_1; \beta) f(x_2; \beta) \kappa(dx_1) \kappa(dx_2) \\
&\quad + \int \int \{q^h(x_1, x_2; \beta) - \tau_1^h\} f(x_1; \beta) \dot{f}(x_2; \beta) \kappa(dx_1) \kappa(dx_2), \tag{A.4}
\end{aligned}$$

where $\dot{f}(x; \beta)$ is the derivative of $f(x; \beta)$ with respect to β . Note that $S_\beta(x; \beta) = \frac{\partial \log[f(x; \beta)]}{\partial \beta} = \frac{\dot{f}(x; \beta)}{f(x; \beta)}$, so the second term on the right-hand side of (A.4) is

$$\begin{aligned}
&\int \int \{q^h(x_1, x_2; \beta) - \tau_1^h\} \dot{f}(x_1; \beta) f(x_2; \beta) \kappa(dx_1) \kappa(dx_2) \\
&= \int \int \{q^h(x_1, x_2; \beta) - \tau_1^h\} f(x_2; \beta) \kappa(dx_2) \dot{f}(x_1; \beta) \kappa(dx_1) \\
&= \int \{q_1^h(\mathcal{O}; \beta) - \tau_1^h\} \dot{f}(x_1; \beta) \kappa(dx_1) = E \{q_1^h(\mathcal{O}; \beta) - \tau_1^h\} S_\beta(\mathcal{O}; \beta).
\end{aligned}$$

Similarly, we can also show that the third term of the right-hand side of (A.4) leads to the same identity. Therefore, rearranging the terms of (A.4), we obtain the result of (A.3) so that the influence function $\phi(\mathcal{O}) = \chi^h(\mathcal{O}) - \prod \{\chi^h(\mathcal{O}) | \Lambda_e\}$. Thus, by the multivariate Pythagorean theorem (Tsiatis, 2006), the variance of $\hat{\tau}_1^h$ is

$$Var(\hat{\tau}_1^h) = Var[\phi(\mathcal{O})] = Var[\chi^h(\mathcal{O}) - \prod \{\chi^h(\mathcal{O}) | \Lambda_e\}] = Var[\chi^h(\mathcal{O})] - Var[\prod \{\chi^h(\mathcal{O}) | \Lambda_e\}]. \tag{A.5}$$

B3. Proof for Theorem 1(c)

In this subsection, we show $var(\chi^h(\mathcal{O}))$ is equal to the variance of unadjusted estimator τ_1^{UNADJ} . To see this point, note that if we let $h(X_i, X_j) = e(X_i)[1 - e(X_j)]$ and $\omega(X_i, X_j) = 1$, then $\hat{\tau}_1^h$ defined in (3.1) of main text is equivalent to τ_1^{UNADJ} defined in (2.3) of main text. From this viewpoint, the unadjusted estimator $\hat{\tau}_1^{\text{UNADJ}}$ also belongs to the same class of balancing weighting estimators as $\hat{\tau}_1^h$. Therefore, by using a similar argument as in Theorem 1(a), the influence function of $\hat{\tau}_1^{\text{UNADJ}}$ is

$$\phi(\mathcal{O})^{\text{UNADJ}} = 2 \{q_1^{\text{UNADJ}}(\mathcal{O}) - \tau_1\},$$

where $q_1^{\text{UNADJ}}(\mathcal{O}) = \mathbb{E} q^{\text{UNADJ}}(\cdot, \mathcal{O}) = \frac{1}{2} \left[\frac{Z}{\pi} g_1(Y) + \frac{1-Z}{1-\pi} g_0(Y) \right]$ with $q^{\text{UNADJ}}(\mathcal{O}_i, \mathcal{O}_j) = \frac{1}{2} [Z_i(1 - Z_j) \mathbb{I}(Y_i > Y_j) + Z_j(1 - Z_i) \mathbb{I}(Y_i < Y_j)] / \left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j) \right]$. Typically, for the unadjusted estimator $\hat{\tau}_1^{\text{UNADJ}}$, $\pi = e(X)$ and $\tau_1^h = \tau_1$, so $2 \{q_1^{\text{UNADJ}}(\mathcal{O}) - \tau_1\} = \chi^h(\mathcal{O})$ and $var(\hat{\tau}_1^{\text{UNADJ}}) = var(\chi^h(\mathcal{O}))$. Therefore, according to the result in (A.5), we know both IPW and OW weighting estimators have no larger variance than the unadjusted estimator. By Theorem 1(c), the asymptotic variance of $n^{1/2} (\hat{\tau}_1^{\text{UNADJ}} - \tau_1)$ can be consistently estimated by

$$n^{-1} \sum_{i=1}^n [2 \{\hat{q}_1^{\text{UNADJ}}(\mathcal{O}_i) - \hat{\tau}_1^{\text{UNADJ}}\}]^{\otimes 2}, \tag{A.6}$$

where $\hat{q}_1^{\text{UNADJ}}(\mathcal{O}_i) = (n-1)^{-1} \sum_{j \neq i}^n q^{\text{UNADJ}}(\mathcal{O}_i, \mathcal{O}_j)$.

Web Appendix C: Concrete expressions for $\tilde{l}_\beta(Z, X; \hat{\beta})$, A_1^{IPW} and A_1^{OW}

C1. Influence function for logistic regression

In this paper, we apply a logistic regression model to describe the propensity score, that is, $e(X_i) = e(X_i; \beta) = \text{Pr}(Z_i = 1|X_i) = \{1 + \exp(-X_i^T \beta)\}^{-1}$ for $i = 1, \dots, n$. Then, the score function of $\hat{\beta}$ is

$$n^{-1/2} \sum_{i=1}^n X_i [Z_i - e(X_i; \hat{\beta})] = 0.$$

By using Taylor expansion, we expand the score equation around the true parameter β , i.e.,

$$\begin{aligned} 0 &= n^{-1/2} \sum_{i=1}^n X_i [Z_i - e(X_i; \hat{\beta})] \\ &= n^{-1/2} \sum_{i=1}^n X_i [Z_i - e(X_i; \beta)] + n^{-1/2} \frac{\partial \{\sum_{i=1}^n X_i [Z_i - e(X_i; \beta)]\}}{\partial \beta} (\hat{\beta} - \beta) + o_p(1) \\ &= n^{-1/2} \sum_{i=1}^n X_i [Z_i - e(X_i; \beta)] - \left[\frac{1}{n} \sum_{i=1}^n X_i X_i^T e(X_i; \beta) \{1 - e(X_i; \beta)\} \right] \sqrt{n} (\hat{\beta} - \beta) + o_p(1), \end{aligned}$$

which leads to

$$\begin{aligned} \sqrt{n} (\hat{\beta} - \beta) &= \left[\frac{1}{n} \sum_{i=1}^n X_i X_i^T e(X_i; \beta) \{1 - e(X_i; \beta)\} \right]^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i [Z_i - e(X_i; \beta)] + o_p(1) \\ &= E_{\beta\beta}^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i [Z_i - e(X_i; \beta)] + o_p(1), \end{aligned}$$

where $E_{\beta\beta} = \frac{1}{n} \sum_{i=1}^n X_i X_i^T e(X_i; \beta) \{1 - e(X_i; \beta)\}$. Thus, according to Tsiatis (2006), the influence function $\tilde{l}_\beta(Z_i, X_i; \hat{\beta})$ is

$$\tilde{l}_\beta(Z_i, X_i; \hat{\beta}) = E_{\hat{\beta}\hat{\beta}}^{-1} X_i [Z_i - e(X_i; \hat{\beta})] = E_{\hat{\beta}\hat{\beta}}^{-1} X_i [Z_i - \hat{e}(X_i)]. \quad (\text{A.7})$$

C2. Expression for A_1^{IPW}

For $q^{\text{IPW}}(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta})$ involving in A_1^{IPW} , conducting derivative of $q^{\text{IPW}}(\mathcal{O}_i, \mathcal{O}_j; \beta)$ with respect to β gives

$$\begin{aligned} &\frac{\partial}{\partial \beta} q^{\text{IPW}}(\mathcal{O}_i, \mathcal{O}_j; \beta) \\ &= \frac{1}{2} \left[\frac{0 - \{e(X_i)(1 - e(X_i))X_i(1 - e(X_j)) + e(X_i)[-e(X_j)(1 - e(X_j))X_j]\}Z_i(1 - Z_j)I(Y_i > Y_j)}{e^2(X_i)(1 - e(X_j))^2} \right. \\ &\quad \left. + \frac{0 - \{e(X_j)(1 - e(X_j))X_j(1 - e(X_i)) + e(X_j)[-e(X_i)(1 - e(X_i))X_i]\}Z_j(1 - Z_i)I(Y_i < Y_j)}{e^2(X_j)(1 - e(X_i))^2} \right] / \\ &\quad \left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n \frac{Z_i(1 - Z_j)}{e(X_i)[1 - e(X_j)]} \right] - \frac{\frac{1}{2} \left[\frac{Z_i(1 - Z_j)I(Y_i > Y_j)}{e(X_i)(1 - e(X_j))} + \frac{Z_j(1 - Z_i)I(Y_i < Y_j)}{e(X_j)(1 - e(X_i))} \right]}{\left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n \frac{Z_i(1 - Z_j)}{e(X_i)[1 - e(X_j)]} \right]^2} \times \\ &\quad \left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n \frac{Z_i(1 - Z_j)[e(X_j)X_j - (1 - e(X_i))X_i]}{e(X_i)[1 - e(X_j)]} \right] \\ &= A^{\text{IPW}}(\mathcal{O}_i, \mathcal{O}_j; \beta) / C^{\text{IPW}} - q^{\text{IPW}}(\mathcal{O}_i, \mathcal{O}_j; \beta) \times B^{\text{IPW}} / C^{\text{IPW}}, \quad (\text{A.8}) \end{aligned}$$

where

$$\begin{aligned}
A^{\text{IPW}}(\mathcal{O}_i, \mathcal{O}_j; \beta) &= \frac{1}{2} \left[\frac{[e(X_j)X_j - (1 - e(X_i))X_i]Z_i(1 - Z_j)I(Y_i > Y_j)}{e(X_i)(1 - e(X_j))} \right. \\
&\quad \left. + \frac{[e(X_i)X_i - (1 - e(X_j))X_j]Z_j(1 - Z_i)I(Y_i < Y_j)}{e(X_j)(1 - e(X_i))} \right], \\
B^{\text{IPW}} &= \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n \frac{Z_i(1 - Z_j)[e(X_j)X_j - (1 - e(X_i))X_i]}{e(X_i)[1 - e(X_j)]}, \\
C^{\text{IPW}} &= \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n \frac{Z_i(1 - Z_j)}{e(X_i)[1 - e(X_j)]}.
\end{aligned}$$

Therefore, A_1^{IPW} can be obtained by putting (A.8) into it.

Remark 1

1. C^{IPW} is a numerical value and B^{IPW} is a vector once we have observed data and obtain $\hat{\beta}$.
2. A_1^{IPW} can be further simplified as followings:

$$\begin{aligned}
A_1^{\text{IPW}} &= \frac{1}{\hat{C}^{\text{IPW}}} \left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n \frac{[\hat{e}(X_j)X_j - (1 - \hat{e}(X_i))X_i]Z_i(1 - Z_j)\mathbb{I}(Y_i > Y_j)}{\hat{e}(X_i)(1 - \hat{e}(X_j))} \right] - \frac{\hat{B}^{\text{IPW}}}{\hat{C}^{\text{IPW}}} \hat{\tau}_1^{\text{IPW}} \\
&= \frac{\sum_{i=1}^n \sum_{j=1}^n [\hat{e}(X_j)X_j - (1 - \hat{e}(X_i))X_i] \omega(X_i, X_j) Z_i(1 - Z_j) \mathbb{I}(Y_i > Y_j)}{\sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j) / [\hat{e}(X_i)(1 - \hat{e}(X_j))]} - \frac{\hat{B}^{\text{IPW}}}{\hat{C}^{\text{IPW}}} \hat{\tau}_1^{\text{IPW}},
\end{aligned}$$

where $\omega(X_i, X_j) = \frac{1}{\hat{e}(X_i)(1 - \hat{e}(X_j))}$.

C3. Expression for A_1^{OW}

Next, we show how to compute $\frac{\partial}{\partial \beta} q^{\text{OW}}(\mathcal{O}_i, \mathcal{O}_j; \beta)$ involving in A_1^{OW} . Conducting derivative of $q^{\text{OW}}(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta})$ with respect to β , we have

$$\begin{aligned}
\frac{\partial}{\partial \beta} q^{\text{OW}}(\mathcal{O}_i, \mathcal{O}_j; \beta) &= \frac{\frac{1}{2}[(1 - e(X_j))X_j - e(X_i)X_i]\omega(X_i, X_j)Z_i(1 - Z_j)I(Y_i > Y_j)}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)(1 - e(X_i))e(X_j)} \\
&\quad + \frac{\frac{1}{2}[(1 - e(X_i))X_i - e(X_j)X_j]\omega(X_j, X_i)Z_j(1 - Z_i)I(Y_i < Y_j)}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)(1 - e(X_i))e(X_j)} \\
&\quad - \frac{\frac{1}{2}[\omega(X_i, X_j)Z_i(1 - Z_j)I(Y_i > Y_j) + \omega(X_j, X_i)Z_j(1 - Z_i)I(Y_i < Y_j)]}{\left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)(1 - e(X_i))e(X_j) \right]^2} \times \\
&\quad \left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)(1 - e(X_i))e(X_j)[(1 - e(X_j))X_j - e(X_i)X_i] \right] \\
&= A^{\text{OW}}(\mathcal{O}_i, \mathcal{O}_j; \beta) / C^{\text{OW}} - q^{\text{OW}}(\mathcal{O}_i, \mathcal{O}_j; \beta) \times B^{\text{OW}} / C^{\text{OW}}, \tag{A.9}
\end{aligned}$$

where

$$\begin{aligned}
A^{\text{OW}}(\mathcal{O}_i, \mathcal{O}_j; \beta) &= \frac{1}{2} \{ [(1 - e(X_j))X_j - e(X_i)X_i] \omega(X_i, X_j) Z_i(1 - Z_j) I(Y_i > Y_j) \\
&\quad + [(1 - e(X_i))X_i - e(X_j)X_j] \omega(X_j, X_i) Z_j(1 - Z_i) I(Y_i < Y_j) \}, \\
B^{\text{OW}} &= \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)(1 - e(X_i))e(X_j)[(1 - e(X_j))X_j - e(X_i)X_i], \\
C^{\text{OW}} &= \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)(1 - e(X_i))e(X_j).
\end{aligned}$$

Therefore, A_1^{OW} can be obtained by putting (A.9) into it.

Remark 2

1. C^{OW} is a numerical value and B^{OW} is a vector once we have observed data and obtain $\hat{\beta}$.
2. A_1^{OW} can be further simplified as followings:

$$\begin{aligned}
A_1^{\text{OW}} &= \frac{1}{\hat{C}^{\text{OW}}} \left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n [(1 - \hat{e}(X_j))X_j - \hat{e}(X_i)X_i] \omega(X_i, X_j) Z_i(1 - Z_j) \mathbb{I}(Y_i > Y_j) \right] - \frac{\hat{B}^{\text{OW}}}{\hat{C}^{\text{OW}}} \hat{\tau}_1^{\text{OW}} \\
&= \frac{\sum_{i=1}^n \sum_{j=1}^n [(1 - \hat{e}(X_j))X_j - \hat{e}(X_i)X_i] \omega(X_i, X_j) Z_i(1 - Z_j) \mathbb{I}(Y_i > Y_j)}{\sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)(1 - \hat{e}(X_i))\hat{e}(X_j)} - \frac{\hat{B}^{\text{OW}}}{\hat{C}^{\text{OW}}} \hat{\tau}_1^{\text{OW}},
\end{aligned}$$

where $\omega(X_i, X_j) = (1 - \hat{e}(X_i))\hat{e}(X_j)$.

Web Appendix D: Concrete expressions for $\tilde{l}_\alpha(Y, X; \hat{\alpha})$, B_1^{AIPW} , B_1^{AOW} , C_1^{AIPW} and C_1^{AOW}

D1. Inference for ordinal logistic regression

To obtain $\hat{\tau}_1^{\text{ah}}$ and its variance, in addition to computing $\hat{\beta}$ and $\hat{e}(\cdot)$, we also need to compute $\hat{\alpha}$ and $\hat{\mu}(\cdot, \cdot)$. In this paper, we apply ordinal logistic regression to model $\mu(\cdot, \cdot; \alpha)$ and use polr program in R package MASS to fit that. Note that in polr program, the ordinal logistic regression model for ordinal outcome with L categories is parameterized as

$$\text{logit}(P(Y \leq l)) = \alpha_{l0} - \xi^T X, \quad (\text{A.10})$$

where $l = 1, 2, \dots, L-1$ and $\xi = (\xi_1, \dots, \xi_p)^T$. Then based on (A.10), we have

$$\log \frac{P(Y \leq l)}{1 - P(Y \leq l)} = \alpha_{l0} - \xi^T X \Leftrightarrow P(Y \leq l) = \frac{\exp(\alpha_{l0} - \xi^T X)}{1 + \exp(\alpha_{l0} - \xi^T X)}. \quad (\text{A.11})$$

Denote $\alpha = (\alpha_{10}, \dots, \alpha_{L-1,0}, \xi^T)^T$, $c_{il} = c_l(X_i; \alpha_{l0}, \xi) = P(Y_i \leq l | X_i) = \frac{\exp(\alpha_{l0} - \xi^T X_i)}{1 + \exp(\alpha_{l0} - \xi^T X_i)}$, $p_{il} = p_l(X_i; \alpha_{l0}, \xi) = P(Y_i = l | X_i)$, $\delta_{il} = 1$ if $Y_i = l$ and $\delta_{il} = 0$ if $Y_i \neq l$, for $i = 1, 2, \dots, n$ and $l = 1, 2, \dots, L$. Then the likelihood and log likelihood function for model (A.11) are

$$L_n(\alpha) = \prod_{i=1}^n \prod_{l=1}^L p_{il}^{\delta_{il}}, \quad \text{and} \quad l_n(\alpha) = \sum_{i=1}^n \sum_{l=1}^L \delta_{il} \log(p_{il}), \quad (\text{A.12})$$

respectively, where (1) $p_{i1} = c_{i1} = P(Y_i \leq 1|X_i) = P(Y_i = 1|X_i)$ if $Y_i = 1$; (2) $p_{il} = c_{il} - c_{i,l-1} = P(Y_i \leq l|X_i) - P(Y_i \leq l-1|X_i)$ if $1 < Y_i = l < L$; (3) $p_{iL} = 1 - c_{i,L-1} = 1 - P(Y_i \leq L-1|X_i)$ if $Y_i = L$.

Note that we consider $L = 3$ in our simulation, so we use $L = 3$ as an example to show how to obtain influence function of α . Influence function of α for other values of L can be obtained similarly. When $L = 3$, then the log-likelihood function (A.12) becomes

$$\begin{aligned} l_n(\alpha) &= \sum_{i=1}^n \sum_{l=1}^3 \delta_{il} \log(p_{il}) = \sum_{i=1}^n [\delta_{i1} \log p_{i1} + \delta_{i2} \log p_{i2} + \delta_{i3} \log p_{i3}] \\ &= \sum_{i=1}^n [\delta_{i1} \log c_{i1} + \delta_{i2} \log(c_{i2} - c_{i1}) + \delta_{i3} \log(1 - c_{i2})], \end{aligned}$$

where $\alpha = (\alpha_{10}, \alpha_{20}, \xi^T)^T$. It is noticed that $c_{i1} = \frac{\exp(\alpha_{10} - \xi^T X_i)}{1 + \exp(\alpha_{10} - \xi^T X_i)}$ and $c_{i2} = \frac{\exp(\alpha_{20} - \xi^T X_i)}{1 + \exp(\alpha_{20} - \xi^T X_i)}$, then $c_{i2} - c_{i1} = \frac{\exp(\alpha_{20} - \xi^T X_i) - \exp(\alpha_{10} - \xi^T X_i)}{(1 + \exp(\alpha_{20} - \xi^T X_i))(1 + \exp(\alpha_{10} - \xi^T X_i))}$. Therefore, the log-likelihood function $l_n(\alpha)$ can be rewritten as

$$\begin{aligned} l_n(\alpha) &= \sum_{i=1}^n \{ \delta_{i1}(\alpha_{10} - \xi^T X_i) - \delta_{i1} \log[1 + \exp(\alpha_{10} - \xi^T X_i)] + \delta_{i2} \log[\exp(\alpha_{20} - \xi^T X_i) - \exp(\alpha_{10} - \xi^T X_i)] \\ &\quad - \delta_{i2} \log[1 + \exp(\alpha_{10} - \xi^T X_i)] - \delta_{i2} \log[1 + \exp(\alpha_{20} - \xi^T X_i)] - \delta_{i3} \log[1 + \exp(\alpha_{20} - \xi^T X_i)] \} \\ &= \sum_{i=1}^n \{ \delta_{i1}(\alpha_{10} - \xi^T X_i) - (\delta_{i1} + \delta_{i2}) \log[1 + \exp(\alpha_{10} - \xi^T X_i)] - (\delta_{i2} + \delta_{i3}) \log[1 + \exp(\alpha_{20} - \xi^T X_i)] \\ &\quad + \delta_{i2} \log[\exp(\alpha_{20} - \xi^T X_i) - \exp(\alpha_{10} - \xi^T X_i)] \}. \end{aligned}$$

Thus, conducting derivative of $l_n(\alpha)$ with respect to α , we have

$$\begin{aligned} \frac{\partial l_n(\alpha)}{\partial \alpha} &= \sum_{i=1}^n \left[\begin{array}{c} \delta_{i1} - (\delta_{i1} + \delta_{i2}) \frac{\exp(\alpha_{10} - \xi^T X_i)}{1 + \exp(\alpha_{10} - \xi^T X_i)} - \delta_{i2} \frac{\exp(\alpha_{10} - \xi^T X_i)}{\exp(\alpha_{20} - \xi^T X_i) - \exp(\alpha_{10} - \xi^T X_i)} \\ - (\delta_{i2} + \delta_{i3}) \frac{\exp(\alpha_{20} - \xi^T X_i)}{1 + \exp(\alpha_{20} - \xi^T X_i)} + \delta_{i2} \frac{\exp(\alpha_{20} - \xi^T X_i)}{\exp(\alpha_{20} - \xi^T X_i) - \exp(\alpha_{10} - \xi^T X_i)} \\ - \delta_{i1} X_i + (\delta_{i1} + \delta_{i2}) \frac{\exp(\alpha_{10} - \xi^T X_i)}{1 + \exp(\alpha_{10} - \xi^T X_i)} X_i + (\delta_{i2} + \delta_{i3}) \frac{\exp(\alpha_{20} - \xi^T X_i)}{1 + \exp(\alpha_{20} - \xi^T X_i)} X_i - \delta_{i2} \frac{\exp(\alpha_{20} - \xi^T X_i) - \exp(\alpha_{10} - \xi^T X_i)}{\exp(\alpha_{20} - \xi^T X_i) - \exp(\alpha_{10} - \xi^T X_i)} X_i \end{array} \right] \\ &= \sum_{i=1}^n \left[\begin{array}{c} \delta_{i1} - (\delta_{i1} + \delta_{i2}) c_{i1} - \delta_{i2} \frac{1}{\exp(\alpha_{20} - \alpha_{10}) - 1} \\ - (\delta_{i2} + \delta_{i3}) c_{i2} + \delta_{i2} \frac{1}{1 - \exp(\alpha_{10} - \alpha_{20})} \\ \{ -(\delta_{i1} + \delta_{i2}) + (\delta_{i1} + \delta_{i2}) c_{i1} + (\delta_{i2} + \delta_{i3}) c_{i2} \} X_i \end{array} \right]. \end{aligned}$$

Then, the score function of α is

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \left[\begin{array}{c} \delta_{i1} - (\delta_{i1} + \delta_{i2}) \hat{c}_{i1} - \delta_{i2} \frac{1}{\exp(\hat{\alpha}_{20} - \hat{\alpha}_{10}) - 1} \\ - (\delta_{i2} + \delta_{i3}) \hat{c}_{i2} + \delta_{i2} \frac{1}{1 - \exp(\hat{\alpha}_{10} - \hat{\alpha}_{20})} \\ \{ -(\delta_{i1} + \delta_{i2}) + (\delta_{i1} + \delta_{i2}) \hat{c}_{i1} + (\delta_{i2} + \delta_{i3}) \hat{c}_{i2} \} X_i \end{array} \right] = 0.$$

Next, we expand the above score equation around the true parameter α , which leads to

$$0 = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left[\begin{array}{c} \delta_{i1} - (\delta_{i1} + \delta_{i2}) c_{i1} - \delta_{i2} \frac{1}{\exp(\alpha_{20} - \alpha_{10}) - 1} \\ - (\delta_{i2} + \delta_{i3}) c_{i2} + \delta_{i2} \frac{1}{1 - \exp(\alpha_{10} - \alpha_{20})} \\ \{ -(\delta_{i1} + \delta_{i2}) + (\delta_{i1} + \delta_{i2}) c_{i1} + (\delta_{i2} + \delta_{i3}) c_{i2} \} X_i \end{array} \right] + \frac{1}{\sqrt{n}} \sum_{i=1}^n \left[\begin{array}{ccc} F_{i11} & F_{i12} & F_{i13} \\ F_{i21} & F_{i22} & F_{i23} \\ F_{i31} & F_{i32} & F_{i33} \end{array} \right] (\hat{\alpha} - \alpha) + o_p(1),$$

where

$$\begin{aligned}
F_{i11} &= -(\delta_{i1} + \delta_{i2})(c_{i1} - c_{i1}^2) - \delta_{i2} \frac{e^{\alpha_{20} - \alpha_{10}}}{(e^{\alpha_{20} - \alpha_{10}} - 1)^2}, \\
F_{i12} &= F_{i21} = \delta_{i2} \frac{e^{\alpha_{20} - \alpha_{10}}}{(e^{\alpha_{20} - \alpha_{10}} - 1)^2}, \\
F_{i13} &= F_{i31}^T = (\delta_{i1} + \delta_{i2})(c_{i1} - c_{i1}^2)X_i^T, \\
F_{i22} &= -(\delta_{i2} + \delta_{i3})(c_{i2} - c_{i2}^2) - \delta_{i2} \frac{e^{\alpha_{10} - \alpha_{20}}}{(1 - e^{\alpha_{10} - \alpha_{20}})^2}, \\
F_{i23} &= F_{i32}^T = (\delta_{i2} + \delta_{i3})(c_{i2} - c_{i2}^2)X_i^T, \\
F_{i33} &= -\{(\delta_{i1} + \delta_{i2})(c_{i1} - c_{i1}^2) + (\delta_{i2} + \delta_{i3})(c_{i2} - c_{i2}^2)\}X_iX_i^T.
\end{aligned}$$

Therefore,

$$\sqrt{n}(\hat{\alpha} - \alpha) = E_{\alpha\alpha}^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \begin{bmatrix} \delta_{i1} - (\delta_{i1} + \delta_{i2})c_{i1} - \delta_{i2} \frac{1}{\exp(\alpha_{20} - \alpha_{10}) - 1} \\ -(\delta_{i2} + \delta_{i3})c_{i2} + \delta_{i2} \frac{1}{1 - \exp(\alpha_{10} - \alpha_{20})} \\ \{-(\delta_{i1} + \delta_{i2}) + (\delta_{i1} + \delta_{i2})c_{i1} + (\delta_{i2} + \delta_{i3})c_{i2}\}X_i \end{bmatrix} + o_p(1),$$

where $E_{\alpha\alpha} = -\frac{1}{n} \sum_{i=1}^n \begin{bmatrix} F_{i11} & F_{i12} & F_{i13} \\ F_{i21} & F_{i22} & F_{i23} \\ F_{i31} & F_{i32} & F_{i33} \end{bmatrix}$. According to Tsiatis (2006), the influence function $\tilde{l}_\alpha(Y_i, X_i; \hat{\alpha})$ is

$$\begin{aligned}
\tilde{l}_\alpha(Y_i, X_i; \hat{\alpha}) &= E_{\hat{\alpha}\hat{\alpha}}^{-1} \begin{bmatrix} \delta_{i1} - (\delta_{i1} + \delta_{i2})\hat{c}_{i1} - \delta_{i2} \frac{1}{\exp(\hat{\alpha}_{20} - \hat{\alpha}_{10}) - 1} \\ -(\delta_{i2} + \delta_{i3})\hat{c}_{i2} + \delta_{i2} \frac{1}{1 - \exp(\hat{\alpha}_{10} - \hat{\alpha}_{20})} \\ \{-(\delta_{i1} + \delta_{i2}) + (\delta_{i1} + \delta_{i2})\hat{c}_{i1} + (\delta_{i2} + \delta_{i3})\hat{c}_{i2}\}X_i \end{bmatrix} \\
&= E_{\hat{\alpha}\hat{\alpha}}^{-1} \begin{bmatrix} \delta_{i1} - (\delta_{i1} + \delta_{i2})c_1(X_i; \hat{\alpha}_{10}, \hat{\xi}) - \delta_{i2} \frac{1}{\exp(\hat{\alpha}_{20} - \hat{\alpha}_{10}) - 1} \\ -(\delta_{i2} + \delta_{i3})c_2(X_i; \hat{\alpha}_{20}, \hat{\xi}) + \delta_{i2} \frac{1}{1 - \exp(\hat{\alpha}_{10} - \hat{\alpha}_{20})} \\ \{-(\delta_{i1} + \delta_{i2}) + (\delta_{i1} + \delta_{i2})c_1(X_i; \hat{\alpha}_{10}, \hat{\xi}) + (\delta_{i2} + \delta_{i3})c_2(X_i; \hat{\alpha}_{20}, \hat{\xi})\}X_i \end{bmatrix} \quad (\text{A.13})
\end{aligned}$$

Specifically, for win statistics of ordinal outcomes, $p_l^z(X; \alpha)$ is denoted as the model based conditional probability of $Y(z) = l$ given X , where $z \in \{0, 1\}$. So, $p_l^z(X; \alpha) = p_{il}$ for z group. Assuming i is from $z = 1$ group and j is from $z = 0$ group, then,

$$\begin{aligned}
\mu(X_i, X_j; \alpha) &= E[\mathbb{I}(Y_i(1) > Y_j(0))] = \sum_{l=1}^3 \sum_{l'=1}^{l-1} p_l^1(X_i; \alpha) p_{l'}^0(X_j; \alpha) \\
&= p_2^1(X_i; \alpha) p_1^0(X_j; \alpha) + p_3^1(X_i; \alpha) p_1^0(X_j; \alpha) + p_3^1(X_i; \alpha) p_2^0(X_j; \alpha) \\
&= (c_{i2} - c_{i1})c_{j1} + (1 - c_{i2})c_{j1} + (1 - c_{i2})(c_{j2} - c_{j1}) \\
&= (1 - c_{i1})c_{j1} + (1 - c_{i2})(c_{j2} - c_{j1}), \\
\mu(X_j, X_i; \alpha) &= E[\mathbb{I}(Y_i(1) < Y_j(0))] = \sum_{l=1}^3 \sum_{l'=l+1}^3 p_l^1(X_i; \alpha) p_{l'}^0(X_j; \alpha) \\
&= p_1^1(X_i; \alpha) p_2^0(X_j; \alpha) + p_1^1(X_i; \alpha) p_3^0(X_j; \alpha) + p_2^1(X_i; \alpha) p_3^0(X_j; \alpha) \\
&= c_{i1}(c_{j2} - c_{j1}) + c_{i1}(1 - c_{j2}) + (c_{i2} - c_{i1})(1 - c_{j2}) \\
&= (1 - c_{j1})c_{i1} + (1 - c_{j2})(c_{i2} - c_{i1}).
\end{aligned}$$

Thus, the derivative of $\mu(X_i, X_j; \alpha)$ and $\mu(X_i, X_j; \alpha)$ with respect to α is

$$\begin{aligned}
\tilde{\mu}(X_i, X_j; \alpha) &= \frac{\partial \mu(X_i, X_j; \alpha)}{\partial \alpha} \\
&= \begin{bmatrix} -c_{i1}(1 - c_{i1})c_{j1} + c_{j1}(1 - c_{j1})(c_{i2} - c_{i1}) \\ -c_{i2}(1 - c_{i2})(c_{j2} - c_{j1}) + (1 - c_{i2})c_{j2}(1 - c_{j2}) \\ \{c_{j1}(c_{i1} - c_{i1}^2) + (c_{i2} - c_{i2}^2)(c_{j2} - c_{j1})\}X_i - \{(c_{i2} - c_{i1})(c_{j1} - c_{j1}^2) + (1 - c_{i2})(c_{j2} - c_{j2}^2)\}X_j \end{bmatrix}, \\
\tilde{\mu}(X_j, X_i; \alpha) &= \frac{\partial \mu(X_j, X_i; \alpha)}{\partial \alpha} \\
&= \begin{bmatrix} -c_{j1}(1 - c_{j1})c_{i1} + c_{i1}(1 - c_{i1})(c_{j2} - c_{j1}) \\ -c_{j2}(1 - c_{j2})(c_{i2} - c_{i1}) + (1 - c_{j2})c_{i2}(1 - c_{i2}) \\ \{c_{i1}(c_{j1} - c_{j1}^2) + (c_{j2} - c_{j2}^2)(c_{i2} - c_{i1})\}X_j - \{(c_{j2} - c_{j1})(c_{i1} - c_{i1}^2) + (1 - c_{j2})(c_{i2} - c_{i2}^2)\}X_i \end{bmatrix},
\end{aligned} \tag{A.14}$$

where c_{i1} , c_{i2} , c_{j1} and c_{j2} are defined in (A.12).

D2. Expressions for B_1^{AIPW} and C_1^{AIPW}

In main text, we have claimed that asymptotic variance of AIPW estimator $\hat{\tau}_1^{\text{AIPW}}$ can be obtained by Theorem 2. Specifically, when $\omega(X_i, X_j) = \frac{1}{\hat{e}(X_i)(1-\hat{e}(X_j))}$, $\hat{\tau}_1^{ah} = \hat{\tau}_1^{\text{AIPW}}$, and relative detailed expressions are as follows:

$$\begin{aligned}
\hat{\tau}_1^{\text{AIPW}} &= \binom{n}{2}^{-1} \sum_{i=1}^n \sum_{j>i}^n q^{\text{AIPW}}(\cdot, \cdot; \hat{\beta}, \hat{\alpha}), \\
q^{\text{AIPW}}(\cdot, \cdot; \hat{\beta}, \hat{\alpha}) &= \frac{\frac{1}{2} \left\{ \frac{Z_i(1-Z_j)[\mathbb{I}(Y_i > Y_j) - \hat{\mu}(X_i, X_j)]}{\hat{e}(X_i)(1-\hat{e}(X_j))} + \frac{Z_j(1-Z_i)[\mathbb{I}(Y_j < Y_i) - \hat{\mu}(X_j, X_i)]}{\hat{e}(X_j)(1-\hat{e}(X_i))} \right\}}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)/[\hat{e}(X_i)(1-\hat{e}(X_j))]} \\
&\quad + \frac{1}{2} [\hat{\mu}(X_i, X_j) + \hat{\mu}(X_j, X_i)], \\
\hat{\mu}(X_i, X_j) &= \mu(X_i, X_j; \hat{\alpha}) = (1 - \hat{c}_{i1})\hat{c}_{j1} + (1 - \hat{c}_{i2})(\hat{c}_{j2} - \hat{c}_{j1}), \\
\hat{\mu}(X_j, X_i) &= \mu(X_j, X_i; \hat{\alpha}) = (1 - \hat{c}_{j1})\hat{c}_{i1} + (1 - \hat{c}_{j2})(\hat{c}_{i2} - \hat{c}_{i1}), \\
\hat{\tau}_1^{\text{AIPW}}(\mathcal{O}_i; \hat{\beta}, \hat{\alpha}) &= (n-1)^{-1} \sum_{j \neq i}^n q^{\text{AIPW}}(\cdot, \cdot; \hat{\beta}, \hat{\alpha}), \\
B_1^{\text{AIPW}} &= \tilde{B}_1^{\text{AIPW}} - \tilde{B}_3^{\text{AIPW}} * \frac{\tilde{B}_2^{\text{AIPW}}}{\tilde{B}_4^{\text{AIPW}}}, \\
\tilde{B}_1^{\text{AIPW}} &= \frac{\sum_{i=1}^n \sum_{j \neq i}^n \frac{[\hat{e}(X_j)X_j - (1-\hat{e}(X_i))X_i]Z_i(1-Z_j)[\mathbb{I}(Y_i > Y_j) - \hat{\mu}(X_i, X_j)]}{[\hat{e}(X_i)(1-\hat{e}(X_j))]}}{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)/[\hat{e}(X_i)(1-\hat{e}(X_j))]}, \\
\tilde{B}_2^{\text{AIPW}} &= \frac{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)[\mathbb{I}(Y_i > Y_j) - \hat{\mu}(X_i, X_j)]/[\hat{e}(X_i)(1-\hat{e}(X_j))]}{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)/[\hat{e}(X_i)(1-\hat{e}(X_j))]}, \\
\tilde{B}_3^{\text{AIPW}} &= \sum_{i=1}^n \sum_{j \neq i}^n \frac{Z_i(1-Z_j)[\hat{e}(X_j)X_j - (1-\hat{e}(X_i))X_i]}{\hat{e}(X_i)[1-\hat{e}(X_j)]}, \\
\tilde{B}_4^{\text{AIPW}} &= \sum_{i=1}^n \sum_{j \neq i}^n \frac{Z_i(1-Z_j)}{\hat{e}(X_i)[1-\hat{e}(X_j)]}, \\
\frac{\partial q^{\text{AIPW}}(\cdot, \cdot; \hat{\beta}, \alpha)}{\partial \alpha} &= \frac{\frac{1}{2} \left\{ \frac{Z_i(1-Z_j)[- \partial \mu(X_i, X_j; \alpha)/\partial \alpha]}{\hat{e}(X_i)(1-\hat{e}(X_j))} + \frac{Z_j(1-Z_i)[- \partial \mu(X_j, X_i; \alpha)/\partial \alpha]}{\hat{e}(X_j)(1-\hat{e}(X_i))} \right\}}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)/[\hat{e}(X_i)(1-\hat{e}(X_j))]} \\
&\quad + \frac{1}{2} \left[\frac{\partial \mu(X_i, X_j; \alpha)}{\partial \alpha} + \frac{\partial \mu(X_j, X_i; \alpha)}{\partial \alpha} \right], \\
C_1^{\text{AIPW}} &= \frac{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)[- \tilde{\mu}(X_i, X_j; \hat{\alpha})]/[\hat{e}(X_i)(1-\hat{e}(X_j))]}{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)/[\hat{e}(X_i)(1-\hat{e}(X_j))]} \\
&\quad + \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n \tilde{\mu}(X_i, X_j; \hat{\alpha}),
\end{aligned}$$

where $\tilde{\mu}(X_i, X_j; \alpha)$ is defined in (A.14).

D3. expressions for B_1^{AOW} and C_1^{AOW}

When $\omega(X_i, X_j) = (1 - \hat{e}(X_i))\hat{e}(X_j)$, then we have $\hat{\tau}_1^{ah} = \hat{\tau}_1^{\text{AOW}}$, and relative detailed expressions are as follows:

$$\begin{aligned}
\hat{\tau}_1^{\text{AOW}} &= \binom{n}{2}^{-1} \sum_{i=1}^n \sum_{j>i}^n q^{\text{AOW}}(\cdot, \cdot; \hat{\beta}, \hat{\alpha}), \\
q^{\text{AOW}}(\cdot, \cdot; \hat{\beta}, \hat{\alpha}) &= \frac{\frac{1}{2}Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)[\mathbb{I}(Y_i > Y_j) - \hat{\mu}(X_i, X_j)]}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)} \\
&\quad + \frac{\frac{1}{2}Z_j(1-Z_i)(1-\hat{e}(X_j))\hat{e}(X_i)[\mathbb{I}(Y_j > Y_i) - \hat{\mu}(X_j, X_i)]}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)} \\
&\quad + \frac{\frac{1}{2}\hat{e}(X_i)(1-\hat{e}(X_i))\hat{e}(X_j)(1-\hat{e}(X_j))[\hat{\mu}(X_i, X_j) + \hat{\mu}(X_j, X_i)]}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n \hat{e}(X_i)\hat{e}(X_j)(1-\hat{e}(X_i))(1-\hat{e}(X_j))}, \\
\hat{q}_1^{\text{AOW}}(\cdot; \hat{\beta}, \hat{\alpha}) &= (n-1)^{-1} \sum_{j \neq i}^n q^{\text{AOW}}(\cdot, \cdot; \hat{\beta}, \hat{\alpha}), \\
B_1^{\text{AOW}} &= \tilde{B}_1^{\text{AOW}} - \tilde{B}_3^{\text{AOW}} * \frac{\tilde{B}_2^{\text{AOW}}}{\tilde{B}_4^{\text{AOW}}} + \tilde{B}_5^{\text{AOW}} - \tilde{B}_7^{\text{AOW}} * \frac{\tilde{B}_6^{\text{AOW}}}{\tilde{B}_8^{\text{AOW}}}, \\
\tilde{B}_1^{\text{AOW}} &= \frac{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)[(1-\hat{e}(X_j))X_j - \hat{e}(X_i)X_i]}{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)} \times [\mathbb{I}(Y_i > Y_j) - \hat{\mu}(X_i, X_j)], \\
\tilde{B}_2^{\text{AOW}} &= \frac{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)[\mathbb{I}(Y_i > Y_j) - \hat{\mu}(X_i, X_j)]}{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)}, \\
\tilde{B}_3^{\text{AOW}} &= \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)[(1-\hat{e}(X_j))X_j - \hat{e}(X_i)X_i], \\
\tilde{B}_4^{\text{AOW}} &= \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j), \\
\tilde{B}_5^{\text{AOW}} &= \frac{\sum_{i=1}^n \sum_{j \neq i}^n h(X_i, X_j)[(1-2\hat{e}(X_i))X_i]\hat{\mu}(X_i, X_j)}{\sum_{i=1}^n \sum_{j \neq i}^n h(X_i, X_j)}, \\
\tilde{B}_6^{\text{AOW}} &= \frac{\sum_{i=1}^n \sum_{j \neq i}^n h(X_i, X_j)\hat{\mu}(X_i, X_j)}{\sum_{i=1}^n \sum_{j \neq i}^n h(X_i, X_j)}, \\
\tilde{B}_7^{\text{AOW}} &= \sum_{i=1}^n \sum_{j \neq i}^n h(X_i, X_j)[(1-2\hat{e}(X_i))X_i + (1-2\hat{e}(X_j))X_j], \\
\tilde{B}_8^{\text{AOW}} &= \sum_{i=1}^n \sum_{j \neq i}^n h(X_i, X_j), \\
\frac{\partial q^{\text{AOW}}(\cdot, \cdot; \hat{\beta}, \alpha)}{\partial \alpha} &= \frac{\frac{1}{2}Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)[- \partial \mu(X_i, X_j; \alpha) / \partial \alpha]}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)} \\
&\quad + \frac{\frac{1}{2}Z_j(1-Z_i)(1-\hat{e}(X_j))\hat{e}(X_i)[- \partial \mu(X_j, X_i; \alpha) / \partial \alpha]}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)} \\
&\quad + \frac{\frac{1}{2}\hat{e}(X_i)(1-\hat{e}(X_i))\hat{e}(X_j)(1-\hat{e}(X_j))[\frac{\partial \mu(X_i, X_j; \alpha)}{\partial \alpha} + \frac{\partial \mu(X_j, X_i; \alpha)}{\partial \alpha}]}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n \hat{e}(X_i)\hat{e}(X_j)(1-\hat{e}(X_i))(1-\hat{e}(X_j))},
\end{aligned}$$

$$C_1^{\text{AOW}} = \frac{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1 - Z_j)(1 - \hat{e}(X_i))\hat{e}(X_j)[- \tilde{\mu}(X_i, X_j; \hat{\alpha})]}{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1 - Z_j)(1 - \hat{e}(X_i))\hat{e}(X_j)} + \frac{\sum_{i=1}^n \sum_{j \neq i}^n h(X_i, X_j)\tilde{\mu}(X_i, X_j; \hat{\alpha})}{\sum_{i=1}^n \sum_{j \neq i}^n \hat{e}(X_i)\hat{e}(X_j)(1 - \hat{e}(X_i))(1 - \hat{e}(X_j))},$$

where $h(X_i, X_j) = \hat{e}(X_i)\hat{e}(X_j)(1 - \hat{e}(X_i))(1 - \hat{e}(X_j))$ and $\tilde{\mu}(X_i, X_j; \alpha)$ is defined in (A.14).

Reference

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Web Appendix E: An Additional Web Table

Table S1: Summary estimated win ratio (WR) and win difference (WD) by unadjusted estimators under two different variance approaches. AESE1 is calculated based on (2.5) of main text and AESE2 is calculated based on (3.11) of main text. CP1 and CP2 are corresponding coverage probability of the 95% confidence intervals. RB: relative bias; MCSD: Monte Carlo standard deviation; AESE: average estimated standard error; CP: coverage probability.

Model	Design	Sample size	Estimand	True	RB(%)	MCSD	AESE1	AESE2	CP1	CP2
Quadratic	Balanced	200	WR	1.453	4.730	0.437	0.426	0.428	0.931	0.932
			WD	0.091	1.652	0.068	0.067	0.068	0.948	0.949
		300	WR	1.453	1.171	0.338	0.334	0.335	0.943	0.944
			WD	0.091	-3.052	0.054	0.055	0.055	0.952	0.952
		400	WR	1.453	2.582	0.298	0.293	0.294	0.946	0.946
			WD	0.091	2.193	0.048	0.048	0.048	0.941	0.941
	Unbalanced	200	WR	1.450	4.186	0.468	0.445	0.447	0.937	0.939
			WD	0.091	0.717	0.075	0.074	0.074	0.940	0.942
		300	WR	1.450	3.008	0.355	0.358	0.359	0.950	0.950
			WD	0.091	2.383	0.059	0.060	0.061	0.949	0.955
		400	WR	1.450	0.096	0.306	0.301	0.302	0.942	0.942
			WD	0.091	-3.963	0.053	0.052	0.053	0.950	0.951
Interaction	Balanced	200	WR	1.663	4.864	0.517	0.489	0.492	0.932	0.933
			WD	0.126	0.766	0.068	0.067	0.068	0.943	0.946
		300	WR	1.663	1.681	0.392	0.384	0.386	0.943	0.943
			WD	0.126	-1.463	0.056	0.055	0.056	0.949	0.950
		400	WR	1.663	2.346	0.341	0.335	0.336	0.948	0.948
			WD	0.126	0.924	0.048	0.048	0.048	0.949	0.949
	Unbalanced	200	WR	1.657	3.853	0.513	0.504	0.508	0.933	0.935
			WD	0.125	-0.005	0.075	0.074	0.075	0.940	0.945
		300	WR	1.657	3.450	0.419	0.409	0.411	0.946	0.947
			WD	0.125	2.109	0.061	0.061	0.061	0.953	0.953
		400	WR	1.657	0.826	0.355	0.345	0.346	0.932	0.932
			WD	0.125	-1.648	0.054	0.053	0.053	0.936	0.940